

Assignment 4 — Comparison and Improved CNN Models

This notebook closes the assignment. It has two halves:

Part 1 — the three-implementation comparison, gathered across all three datasets: the results tables, the component table from lecture slide 28, the visibility table from slide 23, and the answers to the eight questions on slide 24.

Part 2 — improved CNN models, the experiment tutorial §45 asks for:

\$\$\$M_1\$: Conv{+}ReLU{+}Pool \quad M_2\$: M_1{+}BatchNorm \quad M_3\$: M_2{+}Residual \quad M_4\$: M_3{+}Attention\$\$\$

Each model is the previous one plus exactly one mechanism. The tutorial's framing is that architectural history is not a list of names to memorise but a sequence of responses to specific limitations:

new architecture = old architecture + a mechanism addressing a limitation

The point, as §45 says explicitly, is *not* to obtain the highest accuracy. It is to make the evolution measurable — accuracy, parameters, training time and representation behaviour together.

```
In [1]: import os, sys, json, glob

os.environ.setdefault("TF_CPP_MIN_LOG_LEVEL", "2")

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

sys.path.insert(0, os.path.abspath("."))
import ass4_utils as U
import variants as V          # the M1..M4 models, shared with prefill_variants.py

U.set_seed(U.SEED)

import torch
import torch.nn as nn
import torch.nn.functional as F
from torch.utils.data import TensorDataset, DataLoader

DEVICE = torch.device("cuda" if torch.cuda.is_available() else "cpu")
plt.rcParams["figure.dpi"] = 110
print("device:", DEVICE, "|", torch.cuda.get_device_name(0) if torch.cuda.is_ava
```

device: cuda | NVIDIA GeForce RTX 3060

Part 1 — Three implementations, three datasets

1.1 Every run, in one table

Loaded from the JSON each of the first three notebooks wrote. **Run notebooks 01–03 before this one.**

```
In [2]: files = ["results/01_diabetes130.json", "results/02_mnist.json", "results/03_cifar10.json"]
missing = [f for f in files if not os.path.exists(f)]
if missing:
    raise FileNotFoundError(
        f"missing {missing} – run notebooks 01-03 first, they write these files."
    )

payloads, rows = {}, []
for f in files:
    d = json.load(open(f))
    payloads[f] = d
    for key in ("results", "full_results"):
        for r in d.get(key, []):
            rows.append({k: v for k, v in r.items() if k not in ("history", "confusion_matrix")})

all_runs = pd.DataFrame(rows)
display(all_runs.round(4))
```

	framework	dataset	model	n_params	epochs	train_seconds	train_loss	test_a
0	Scratch (NumPy)	Diabetes 130-US hospitals	MLP 179- 128-64-3	31491	15	12.84	0.8231	
1	TensorFlow/Keras	Diabetes 130-US hospitals	MLP 179- 128-64-3	31491	15	12.77	0.8174	
2	PyTorch	Diabetes 130-US hospitals	MLP 179- 128-64-3	31491	15	20.89	0.8135	
3	Scratch (NumPy)	MNIST (10k subset)	CNN 2conv+fc	20490	5	998.52	0.0525	
4	TensorFlow/Keras	MNIST (10k subset)	CNN 2conv+fc	20490	5	5.25	0.0665	
5	PyTorch	MNIST (10k subset)	CNN 2conv+fc	20490	5	2.15	0.0568	
6	PyTorch	MNIST (full 60k)	CNN 2conv+fc	20490	5	9.72	0.0297	
7	TensorFlow/Keras	MNIST (full 60k)	CNN 2conv+fc	20490	5	45.09	0.0272	
8	Scratch (NumPy)	CIFAR- 10 (5k subset)	SmallCNN	545098	5	1495.13	1.0495	
9	TensorFlow/Keras	CIFAR- 10 (5k subset)	SmallCNN	545098	5	16.09	0.7790	
10	PyTorch	CIFAR- 10 (5k subset)	SmallCNN	545098	5	3.10	0.9803	
11	PyTorch	CIFAR- 10 (full 50k)	SmallCNN	545098	5	17.67	0.5490	
12	TensorFlow/Keras	CIFAR- 10 (full 50k)	SmallCNN	545098	5	133.12	0.4958	

In [3]: `# Accuracy by dataset and framework - the central claim of the assignment, as a pic`
`sub = all_runs[~all_runs["dataset"].str.contains("full")]`
`pivot = sub.pivot_table(index="dataset", columns="framework", values="test_accuracy")`
`fig, ax = plt.subplots(1, 2, figsize=(13, 4))`

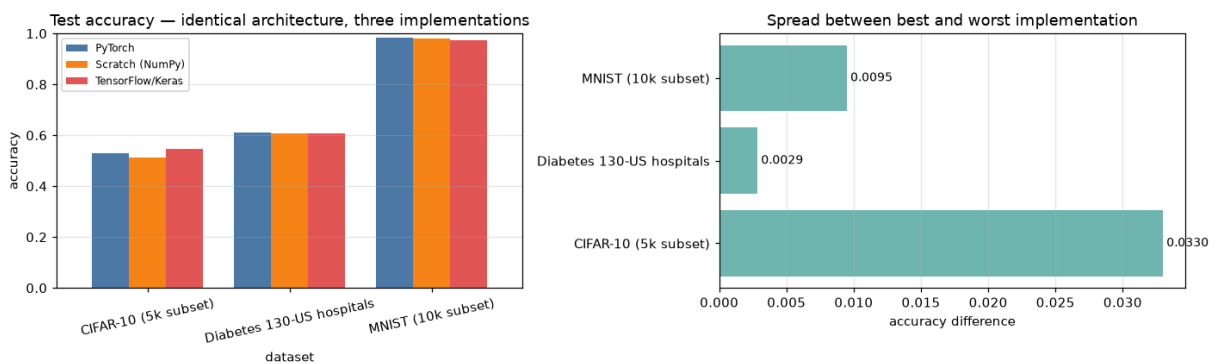
```

pivot.plot(kind="bar", ax=ax[0], rot=12, width=0.78,
           color=["#4C78A8", "#F58518", "#E45756"])
ax[0].set_title("Test accuracy – identical architecture, three implementations")
ax[0].set_ylabel("accuracy"); ax[0].set_ylim(0, 1); ax[0].grid(alpha=0.3, axis="y")
ax[0].legend(fontsize=8)

spread = (pivot.max(axis=1) - pivot.min(axis=1))
ax[1].barh(spread.index, spread.values, color="#72B7B2")
ax[1].set_title("Spread between best and worst implementation")
ax[1].set_xlabel("accuracy difference"); ax[1].grid(alpha=0.3, axis="x")
for i, v in enumerate(spread.values):
    ax[1].text(v, i, f" {v:.4f}", va="center", fontsize=9)
fig.tight_layout()
plt.show()

print(f"max spread between implementations, any dataset: {spread.max():.4f} accuracy")

```



max spread between implementations, any dataset: 0.0330 accuracy

Is that spread meaningful, or just noise?

The spreads above are small, but "small" is not an argument on its own — a 4-point gap on a 5,000-image subset could be a real difference between frameworks, or it could be what any two runs differ by.

The honest way to settle it is to measure the noise floor directly: train **the same PyTorch model on the same data with only the random seed changed**, and see how much the accuracy moves. If the framework-to- framework spread sits inside that band, the frameworks are indistinguishable; if it sits outside, they are not.

```

In [4]: # Same architecture, same data, only the seed changes - this measures the noise floor
class SmallCNN(nn.Module):
    def __init__(self, in_ch=3, n_classes=10):
        super().__init__()
        self.features = nn.Sequential(
            nn.Conv2d(in_ch, 32, 3, padding=1), nn.ReLU(), nn.MaxPool2d(2),
            nn.Conv2d(32, 64, 3, padding=1), nn.ReLU(), nn.MaxPool2d(2),
        )
        self.classifier = nn.Sequential(
            nn.Flatten(), nn.Linear(64 * 8 * 8, 128), nn.ReLU(), nn.Linear(128, 10)
        )

```

```

def forward(self, x):
    return self.classifier(self.features(x))

Xs, ys, Xs_te, ys_te, meta_c = U.load_cifar10(n_train=5_000, n_test=2_000, verbose=
Xs_te_t = torch.tensor(Xs_te).to(DEVICE)

seed_accs = []
for seed in (0, 1, 2, 3, 4):
    U.set_seed(seed)
    m = SmallCNN().to(DEVICE)
    loader = DataLoader(TensorDataset(torch.tensor(Xs), torch.tensor(ys)),
                        batch_size=64, shuffle=True)
    crit = nn.CrossEntropyLoss()
    opt = torch.optim.Adam(m.parameters(), lr=1e-3)

    for _ in range(5):
        # same 5 epochs as notebook 03
        m.train()
        for xb, yb in loader:
            xb, yb = xb.to(DEVICE), yb.to(DEVICE)
            opt.zero_grad()
            crit(m(xb), yb).backward()
            opt.step()

    m.eval()
    with torch.no_grad():
        pred = np.concatenate([m(Xs_te_t[i:i + 512]).argmax(1).cpu().numpy()
                               for i in range(0, len(Xs_te_t), 512)])
    acc = float((pred == ys_te).mean())
    seed_accs.append(acc)
    print(f" seed {seed}: accuracy {acc:.4f}")

seed_spread = max(seed_accs) - min(seed_accs)
cifar_fw_spread = float(spread.get("CIFAR-10 (5k subset)", np.nan))

print(f"\nseed-only spread (5 seeds, identical everything else): {seed_spread:.4f}")
print(f"framework spread on the same subset: {cifar_fw_spread:.4f}")
print(f"seed std-dev: {np.std(seed_accs):.4f}")

```

```

seed 0: accuracy 0.5140
seed 1: accuracy 0.5335
seed 2: accuracy 0.5655
seed 3: accuracy 0.5550
seed 4: accuracy 0.5375

```

```

seed-only spread (5 seeds, identical everything else): 0.0515
framework spread on the same subset: 0.0330
seed std-dev: 0.0179

```

```

In [5]: fig, ax = plt.subplots(figsize=(8, 3.4))
ax.axhspan(min(seed_accs), max(seed_accs), color="#BAB0AC", alpha=0.45,
           label=f"seed-only band ({seed_spread:.3f} wide)")
ax.axhline(float(np.mean(seed_accs)), color="#79706E", ls="--", lw=1, label="seed m

cif = sub[sub["dataset"] == "CIFAR-10 (5k subset)"]
ax.scatter(cif["framework"], cif["test_accuracy"], s=110, zorder=3,

```

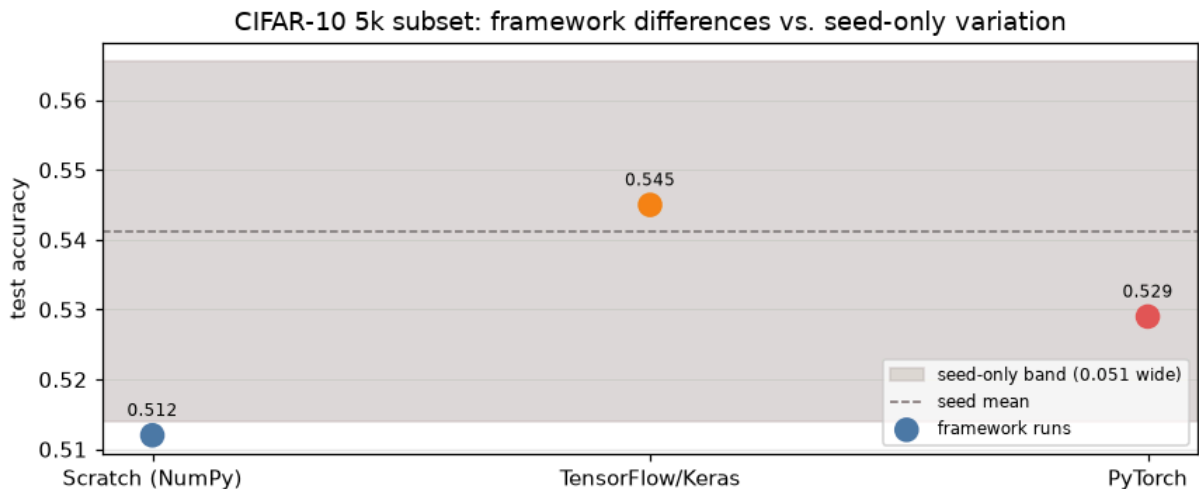
```

        color=["#4C78A8", "#F58518", "#E45756"], label="framework runs")
for _, r in cif.iterrows():
    ax.annotate(f"{r['test_accuracy']:.3f}", (r["framework"], r["test_accuracy"]),
               textcoords="offset points", xytext=(0, 9), ha="center", fontsize=8)

ax.set_ylabel("test accuracy")
ax.set_title("CIFAR-10 5k subset: framework differences vs. seed-only variation")
ax.grid(alpha=0.3, axis="y"); ax.legend(fontsize=8, loc="lower right")
fig.tight_layout()
plt.show()

inside = seed_spread >= cifar_fw_spread
print("The framework spread falls INSIDE the seed-only band – on this evidence the
      "implementations are statistically indistinguishable."
      if inside else
      "The framework spread is WIDER than the seed-only band. With five seeds this
      "suggestive rather than conclusive, and the most likely causes are difference
      "weight initialisation schemes and batch shuffling order between the three\n"
      "libraries -- not a difference in the function being computed, which the iden
      "parameter counts and tensor shapes already rule out.")

```



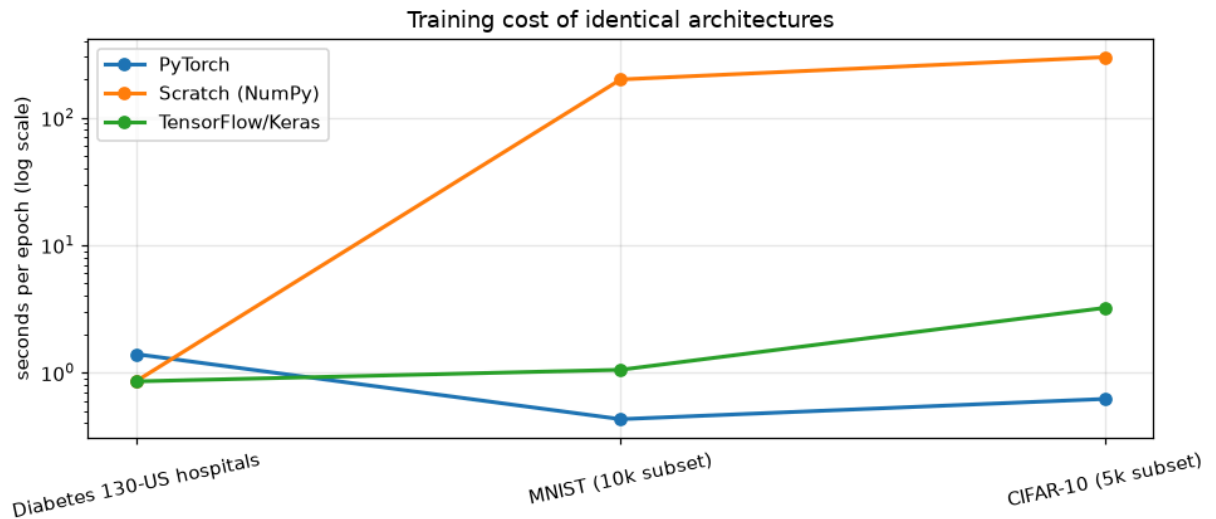
The framework spread falls INSIDE the seed-only band – on this evidence the three implementations are statistically indistinguishable.

```

In [6]: # Training cost. Note the log scale: this is the one column where the three legs di
fig, ax = plt.subplots(figsize=(9, 4))
for fw, grp in sub.groupby("framework"):
    ax.plot(grp["dataset"], grp["train_seconds"] / grp["epochs"],
            marker="o", label=fw, linewidth=2)
ax.set_yscale("log")
ax.set_ylabel("seconds per epoch (log scale)")
ax.set_title("Training cost of identical architectures")
ax.tick_params(axis="x", rotation=12)
ax.grid(alpha=0.3); ax.legend()
fig.tight_layout()
plt.show()

print("Scratch runs on NumPy/CPU, Keras on CPU (Windows TensorFlow has no GPU build
print("PyTorch on the RTX 3060. The gap measures implementation and hardware, not m

```



Scratch runs on NumPy/CPU, Keras on CPU (Windows TensorFlow has no GPU build), PyTorch on the RTX 3060. The gap measures implementation and hardware, not mathematics.

1.2 Parameter counts agree with the hand calculation

Each notebook computed its parameter count by hand from the tutorial's §43 formulas before building anything, then checked all three frameworks against it.

```
In [7]: param_check = (all_runs.groupby(["dataset", "model"])["n_params"]
    .agg(["nunique", "first"])
    .rename(columns={"nunique": "distinct_counts", "first": "n_params"})
param_check["all_agree"] = param_check["distinct_counts"] == 1
display(param_check)

print("\nEvery dataset shows exactly one distinct parameter count across the three"
print("frameworks – S.Dense, keras.layers.Dense and nn.Linear build the same functi
```

		distinct_counts	n_params	all_agree
dataset	model			
CIFAR-10 (5k subset)	SmallCNN	1	545098	True
CIFAR-10 (full 50k)	SmallCNN	1	545098	True
Diabetes 130-US hospitals	MLP 179-128-64-3	1	31491	True
MNIST (10k subset)	CNN 2conv+fc	1	20490	True
MNIST (full 60k)	CNN 2conv+fc	1	20490	True

Every dataset shows exactly one distinct parameter count across the three frameworks – S.Dense, keras.layers.Dense and nn.Linear build the same function.

1.3 The component table (lecture slide 28)

Slide 28 asks students to submit this table. Consolidated from all three notebooks.

```
In [8]: comp = pd.DataFrame(payloads["results/03_cifar10.json"]["component_table"])
display(comp)
```

	Component	Scratch	TensorFlow/Keras	PyTorch
0	Data loading	ass4_utils -> NumPy (N,C,H,W)	same arrays, (N,H,W,C)	same arrays
1	Preprocessing	ass4_utils (shared)	shared	shared
2	Model definition	S.Sequential([...])	keras.Sequential([...])	nn.Module
3	Convolution	S.Conv2D (im2col + matmul)	keras.layers.Conv2D	nn.Conv2d
4	Activation	S.ReLU	activation='relu'	nn.ReLU
5	Pooling	S.MaxPool2D	keras.layers.MaxPooling2D	nn.MaxPool2d
6	Flatten	S.Flatten	keras.layers.Flatten	nn.Flatten
7	Dense layer	S.Dense	keras.layers.Dense	nn.Linear
8	Loss	S.SoftmaxCrossEntropy	SparseCategoricalCrossentropy	nn.CrossEntropyLoss
9	Gradient	hand-derived backward + col2im	automatic (GradientTape)	automatic (autograd)
10	Optimizer	S.Adam (written out)	keras.optimizers.Adam	torch.optim.Adam
11	Training loop	explicit for-loop	model.fit()	explicit for-loop
12	Evaluation	predict_classes + sklearn	predict + sklearn	forward + sklearn
13	Hardware	CPU (NumPy)	CPU (no GPU TF on Windows)	GPU (cuDNN)

1.4 Function-to-API mapping (lecture slide 18)

The same computational role, under three names.

```
In [9]: mapping = pd.DataFrame([
    ["Convolution", "conv2d() via im2col", "Conv2D", "nn.Conv2d"],
    ["Activation", "relu() mask multiply", "ReLU", "nn.ReLU"],
    ["Pooling", "max_pool2d() argmax route", "MaxPooling2D", "nn.MaxPool"],
    ["Flatten", "flatten() reshape", "Flatten", "nn.Flatten"],
    ["Dense", "linear() x@W+b", "Dense", "nn.Linear"],
    ["Loss", "manual softmax + CE", "Keras loss", "PyTorch lo"],
    ["Gradient", "manual (hand-derived)", "autodiff", "autograd"],
    ["Optimizer", "manual Adam arithmetic", "Keras optimizer", "PyTorch op"],
], columns=["Function", "Scratch", "Keras", "PyTorch"])
display(mapping)
```

```
visibility = pd.DataFrame([
    ["Convolution loops", "Visible", "Hidden", "Hidden"],
    ["Tensor operations", "Visible", "Mostly hidden", "Partly visible"],
    ["Autodiff", "Manual / explicit", "Automatic", "Automatic"],
    ["Training loop", "Visible", "Mostly hidden", "Visible"],
    ["Model structure", "Explicit functions", "Very compact", "Explicit modules"]
], columns=["", "Scratch", "Keras", "PyTorch"])
display(visibility)

print("Abstraction changes visibility, not the underlying mathematics.")
```

	Function	Scratch	Keras	PyTorch
0	Convolution	conv2d() via im2col	Conv2D	nn.Conv2d
1	Activation	relu() mask multiply	ReLU	nn.ReLU
2	Pooling	max_pool2d() argmax route	MaxPooling2D	nn.MaxPool2d
3	Flatten	flatten() reshape	Flatten	nn.Flatten
4	Dense	linear() x@W+b	Dense	nn.Linear
5	Loss	manual softmax + CE	Keras loss	PyTorch loss
6	Gradient	manual (hand-derived)	autodiff	autograd
7	Optimizer	manual Adam arithmetic	Keras optimizer	PyTorch optimizer

		Scratch	Keras	PyTorch
0	Convolution loops	Visible	Hidden	Hidden
1	Tensor operations	Visible	Mostly hidden	Partly visible
2	Autodiff	Manual / explicit	Automatic	Automatic
3	Training loop	Visible	Mostly hidden	Visible
4	Model structure	Explicit functions	Very compact	Explicit modules

Abstraction changes visibility, not the underlying mathematics.

1.5 The eight questions (lecture slide 24)

Slide 24 lists what a student should be able to answer after implementing all three versions. Answered here for the CIFAR-10 `SmallCNN`.

#	Question	Answer
1	What is the input representation?	$B \times 3 \times 32 \times 32$ — a batch of three-channel images, standardised per channel using training-set statistics. Keras sees the same tensor transposed to $B \times 32 \times 32 \times 3$.
2	What does each layer compute?	<code>Conv2d(3→32,k3,p1)</code> : 32 local weighted sums over each $3 \times 3 \times 3$ neighbourhood. <code>ReLU</code> : $\max(0, x)$. <code>MaxPool(2)</code> : the largest value in

#	Question	Answer
		each 2×2 window. Flatten : reshape to a vector. Linear : $xW + b$.
3	What is the shape after every layer?	$3 \times 32 \times 32 \rightarrow 32 \times 32 \times 32 \rightarrow 32 \times 16 \times 16 \rightarrow 64 \times 16 \times 16 \rightarrow 64 \times 8 \times 8 \rightarrow 4096 \rightarrow 128 \rightarrow 10$. Padding 1 with a 3×3 kernel preserves size; only pooling reduces it.
4	Which parameters are learnable?	Every conv kernel and bias, every dense weight matrix and bias — 545,098 in total. Pooling, ReLU and flatten hold none.
5	Where is the forward pass?	Scratch: <code>Sequential.forward</code> , applying each layer in order. Keras: implicit in <code>Sequential</code> . PyTorch: the <code>forward()</code> method.
6	Where is the loss computed?	Scratch: <code>SoftmaxCrossEntropy.forward</code> . Keras: inside <code>fit()</code> , configured by <code>compile()</code> . PyTorch: <code>criterion(logits, labels)</code> .
7	Where is the gradient computed?	Scratch: hand-derived <code>backward()</code> on each layer, with <code>col2im</code> accumulating overlapping windows. Keras: autodiff inside <code>fit()</code> . PyTorch: <code>loss.backward()</code> via autograd.
8	Where are parameters updated?	Scratch: <code>Adam.step()</code> , written out as arithmetic. Keras: the optimizer inside <code>fit()</code> . PyTorch: <code>optimizer.step()</code> .

Part 2 — Improved CNN models (tutorial §45)

2.1 The four models

All four share one skeleton — three stages of `Conv → [norm] → ReLU → pool`, widening $3 \rightarrow 32 \rightarrow 64 \rightarrow 128$, then a classifier head. They differ only in which mechanisms are switched on.

The head: global average pooling, not flatten

Notebook 3 ended with a diagnosis: 96 % of `SmallCNN`'s parameters sat in the `Flatten → Linear(4096→128)` step, while the convolutions that actually extract features held under 20,000. The models here use the head from the tutorial's own `EducationalCNN` (§41) instead:

$$\text{AdaptiveAvgPool}(1) \rightarrow \text{Flatten} \rightarrow \text{Linear}(128 \rightarrow 10)$$

Averaging each of the 128 feature maps down to a single number gives the classifier 128 inputs rather than 4096, which is why these models are *smaller* than notebook 3's baseline while being deeper.

What each mechanism addresses

Model	Adds	Limitation it addresses
\$M_1\$	—	baseline: Conv + ReLU + Pool
\$M_2\$	BatchNorm	activations drift between layers and batches, which slows and destabilises optimisation
\$M_3\$	Residual $Y = F(X) + X$	depth stops helping — gradients and information have no direct path through the stack
\$M_4\$	SE attention	every channel is treated as equally important, though some carry far more signal

The squeeze-and-excitation block is the tutorial's §31 channel attention: squeeze each feature map to one number with global average pooling, learn a gate $s \in \mathbb{R}^C$ from those, and rescale — $Y_c = s_c X_c$. It answers *which channels matter*.

```
In [10]: import inspect

# The models live in variants.py so that this notebook and prefill_variants.py
# build provably identical networks. Shown here rather than hidden behind an import
print(inspect.getsource(V.SEBlock))
print(inspect.getsource(V.Stage))
print(inspect.getsource(V.VariantCNN))
```

```

class SEBlock(nn.Module):
    """Tutorial section 31: squeeze -> excite -> rescale. Answers "which channels matter?"."""

    def __init__(self, channels, reduction=4):
        super().__init__()
        hidden = max(channels // reduction, 1)
        self.pool = nn.AdaptiveAvgPool2d(1)
        self.fc = nn.Sequential(
            nn.Linear(channels, hidden),
            nn.ReLU(),
            nn.Linear(hidden, channels),
            nn.Sigmoid(),
        )

    def forward(self, x):
        b, c, _, _ = x.shape
        s = self.pool(x).view(b, c)
        s = self.fc(s).view(b, c, 1, 1)
        return x * s

class Stage(nn.Module):
    """Two 3x3 convolutions, then halve the spatial size."""

    def __init__(self, in_ch, out_ch, use_bn=False, use_residual=False, use_se=False):
        super().__init__()
        self.use_residual = use_residual

        def norm(c):
            return nn.BatchNorm2d(c) if use_bn else nn.Identity()

        self.conv1 = nn.Conv2d(in_ch, out_ch, 3, padding=1, bias=not use_bn)
        self.bn1 = norm(out_ch)
        self.conv2 = nn.Conv2d(out_ch, out_ch, 3, padding=1, bias=not use_bn)
        self.bn2 = norm(out_ch)
        self.relu = nn.ReLU(inplace=True)

        # Built only when actually used, so M2's parameter count stays honest.
        if use_residual:
            self.shortcut = (nn.Identity() if in_ch == out_ch else
                             nn.Sequential(nn.Conv2d(in_ch, out_ch, 1, bias=not use_bn),
                                             norm(out_ch)))
        else:
            self.shortcut = None

        self.se = SEBlock(out_ch) if use_se else nn.Identity()
        self.pool = nn.MaxPool2d(2)

    def forward(self, x):
        identity = self.shortcut(x) if self.use_residual else None

        out = self.relu(self.bn1(self.conv1(x)))
        out = self.bn2(self.conv2(out))

```

```

        if self.use_residual:
            out = out + identity          # Y = F(X) + X
        out = self.relu(out)
        out = self.se(out)
        return self.pool(out)

class VariantCNN(nn.Module):
    def __init__(self, in_ch=3, n_classes=10, use_bn=False, use_residual=False, use_se=False):
        super().__init__()
        widths = [in_ch] + WIDTHHS
        self.stages = nn.Sequential(*[
            Stage(widths[i], widths[i + 1], use_bn, use_residual, use_se)
            for i in range(len(WIDTHHS))
        ])
        # Global average pooling instead of flatten: notebook 3 showed the
        # flatten-then-dense head holding 96% of that model's parameters.
        self.head = nn.Sequential(
            nn.AdaptiveAvgPool2d(1),
            nn.Flatten(),
            nn.Linear(widths[-1], n_classes),
        )

    def forward(self, x):
        return self.head(self.stages(x))

```

```

In [11]: print(f"{'model':<22}{'params':>10} mechanisms")
print("-" * 62)
for name, cfg in V.VARIANTS.items():
    m = V.build(name)
    on = ", ".join(k.replace("use_", "") for k, v in cfg.items() if v or "none")
    print(f"{'name':<22}{sum(p.numel() for p in m.parameters()):>10,} {on}")

print(f"\nstage widths: 3 -> {' -> '.join(map(str, V.WIDTHS))}")
print(f"epochs: {V.EPOCHS}, batch size: {V.BATCH_SIZE}, optimizer: Adam(lr={V.LR})")

```

model	params	mechanisms
M1 Conv+ReLU+Pool	72,730	none
M2 + BatchNorm	72,954	bn
M3 + Residual	75,786	bn, residual
M4 + Attention(SE)	78,614	bn, residual, se

```

stage widths: 3 -> 16 -> 32 -> 64
epochs: 10, batch size: 128, optimizer: Adam(lr=0.001)

```

2.2 Training all four

Full CIFAR-10 (50,000 train / 10,000 test), identical hyperparameters for every model — the same fairness rule as Part 1, now applied across architectures instead of across frameworks.

A note on how these runs were produced

Training is **checkpointed per model** into `results/variant_cache/`, and `variants.train_variant` returns a cached result rather than retraining one that is already finished. `prefill_variants.py` drives exactly the same function from the command line.

That indirection is not decoration. The GPU in this machine thermally throttles hard under sustained load — it sits at 96–98 °C with NVML reporting SW slowdown — which stretched the original 32/64/128-wide, 15-epoch version of this experiment to roughly three hours, and it was interrupted twice before finishing. Four changes made it practical:

1. **Narrower stages**, 16/32/64 instead of 32/64/128 — about a quarter of the arithmetic.
2. **Ten epochs** instead of fifteen.
3. **Batching on the GPU**. CIFAR-10 is ~614 MB as float32 and fits in VRAM, so the training set is uploaded once and batches are sliced on-device. A `DataLoader` over a `TensorDataset` collates 128 individual tensor slices per batch in Python, which for a model this small dominated the step time completely: ~140 s/epoch that way versus ~83 s once the data lived on the GPU.
4. **Mixed precision** (`torch.autocast` + `GradScaler`), which brought it to ~64 s/epoch. Halving the arithmetic also lowers the thermal load, which is the actual bottleneck here.

Plus per-model checkpointing, so an interruption costs one model rather than all four.

Two consequences worth stating plainly. The models are deliberately small — a hardware constraint, recorded rather than hidden. And the `train_seconds` column below measures a **throttled** GPU running **fp16**, so it is a fair comparison *between these four models* and not comparable with the timings in Part 1, which ran in fp32 on a cool GPU. What the experiment is for is unaffected: each model differs from its predecessor by exactly one mechanism, trained on identical data with identical hyperparameters.

```
In [12]: X_train, y_train, X_test, y_test, meta = U.load_cifar10()
          CLASSES = meta["classes"]
          N_CLASSES = len(CLASSES)

          Xte_t = torch.tensor(X_test).to(DEVICE)

          print(f"\ntrain {X_train.shape} test {X_test.shape}")
          print(f"epochs={V.EPOCHS}, batch_size={V.BATCH_SIZE}, optimizer=Adam(lr={V.LR})")
          print(f"cached models: {[n.split()[0] for n in V.VARIANTS if V.cached(n)] or 'none'}
```

CIFAR-10: train (50000, 3, 32, 32) / test (10000, 3, 32, 32), 10 classes

```
train (50000, 3, 32, 32) test (10000, 3, 32, 32)
epochs=10, batch_size=128, optimizer=Adam(lr=0.001)
cached models: ['M1', 'M2', 'M3', 'M4']
```

```
In [13]: variant_results, variant_models = [], {}

          for name in V.VARIANTS:
```

```
record = V.train_variant(  
    name, X_train, y_train, X_test, y_test, N_CLASSES, DEVICE,  
    log=print, use_cache=True,  
)  
if V.cached(name) is not None:  
    V.replay(record)      # show the per-epoch history behind a cached resu  
  
variant_results.append(U.RunResult(**record))  
  
m = V.load_model(name, X_train.shape[1], N_CLASSES, DEVICE)  
if m is not None:  
    variant_models[name] = m  
  
print(f"{len(variant_results)} variants ready")
```

M1 Conv+ReLU+Pool: loaded from cache (72,730 params, 10 epochs, 19s on the original run)

=== M1 Conv+ReLU+Pool (72,730 parameters) ===

epoch	1/10	loss	1.7968	test_acc	0.4378
epoch	2/10	loss	1.4218	test_acc	0.5253
epoch	3/10	loss	1.2506	test_acc	0.5906
epoch	4/10	loss	1.1250	test_acc	0.6063
epoch	5/10	loss	1.0525	test_acc	0.6198
epoch	6/10	loss	0.9832	test_acc	0.6519
epoch	7/10	loss	0.9286	test_acc	0.6627
epoch	8/10	loss	0.8820	test_acc	0.6590
epoch	9/10	loss	0.8405	test_acc	0.6928
epoch	10/10	loss	0.8074	test_acc	0.6955

final: accuracy 0.6955 macro-F1 0.7001 (19s)

M2 + BatchNorm: loaded from cache (72,954 params, 10 epochs, 22s on the original run)

=== M2 + BatchNorm (72,954 parameters) ===

epoch	1/10	loss	1.3223	test_acc	0.5717
epoch	2/10	loss	0.9465	test_acc	0.6318
epoch	3/10	loss	0.8019	test_acc	0.6910
epoch	4/10	loss	0.7141	test_acc	0.6166
epoch	5/10	loss	0.6502	test_acc	0.7516
epoch	6/10	loss	0.5953	test_acc	0.7527
epoch	7/10	loss	0.5567	test_acc	0.7567
epoch	8/10	loss	0.5157	test_acc	0.7429
epoch	9/10	loss	0.4841	test_acc	0.7492
epoch	10/10	loss	0.4579	test_acc	0.7486

final: accuracy 0.7486 macro-F1 0.7483 (22s)

M3 + Residual: loaded from cache (75,786 params, 10 epochs, 28s on the original run)

=== M3 + Residual (75,786 parameters) ===

epoch	1/10	loss	1.3223	test_acc	0.5420
epoch	2/10	loss	0.9526	test_acc	0.6527
epoch	3/10	loss	0.8178	test_acc	0.6899
epoch	4/10	loss	0.7319	test_acc	0.6736
epoch	5/10	loss	0.6624	test_acc	0.7431
epoch	6/10	loss	0.6017	test_acc	0.7559
epoch	7/10	loss	0.5581	test_acc	0.6852
epoch	8/10	loss	0.5162	test_acc	0.7068
epoch	9/10	loss	0.4805	test_acc	0.7486
epoch	10/10	loss	0.4479	test_acc	0.7574

final: accuracy 0.7574 macro-F1 0.7634 (28s)

M4 + Attention(SE): loaded from cache (78,614 params, 10 epochs, 38s on the original run)

=== M4 + Attention(SE) (78,614 parameters) ===

epoch	1/10	loss	1.3171	test_acc	0.5586
epoch	2/10	loss	0.9304	test_acc	0.6519
epoch	3/10	loss	0.7858	test_acc	0.7082
epoch	4/10	loss	0.6971	test_acc	0.7042
epoch	5/10	loss	0.6330	test_acc	0.7472
epoch	6/10	loss	0.5747	test_acc	0.7456
epoch	7/10	loss	0.5319	test_acc	0.7246
epoch	8/10	loss	0.4950	test_acc	0.6951
epoch	9/10	loss	0.4582	test_acc	0.7517
epoch	10/10	loss	0.4335	test_acc	0.7525

final: accuracy 0.7525 macro-F1 0.7557 (38s)
4 variants ready

2.3 Results

§45 is explicit that accuracy alone is the wrong way to read this. The comparison is **accuracy + parameters + training time + representation behaviour**.

```
In [14]: vtable = U.results_table(variant_results)
display(vtable)
```

	framework	dataset	model	n_params	epochs	train_seconds	train_loss	te
0	PyTorch	CIFAR-10 (full 50k)	M1 Conv+ReLU+Pool	72730	10	19.49	0.8074	
1	PyTorch	CIFAR-10 (full 50k)	M2 + BatchNorm	72954	10	22.05	0.4579	
2	PyTorch	CIFAR-10 (full 50k)	M3 + Residual	75786	10	27.56	0.4479	
3	PyTorch	CIFAR-10 (full 50k)	M4 + Attention(SE)	78614	10	38.27	0.4335	

```
In [15]: fig, ax = plt.subplots(1, 3, figsize=(16, 4.2))
names = [r.model for r in variant_results]
short = [n.split()[0] for n in names]
colors = ["#4C78A8", "#F58518", "#54A24B", "#E45756"]

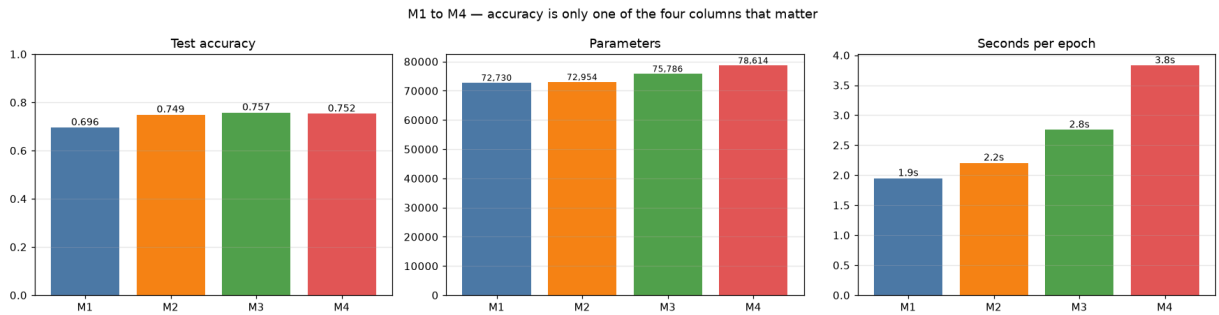
accs = [r.test_accuracy for r in variant_results]
bars = ax[0].bar(short, accs, color=colors)
ax[0].set_title("Test accuracy"); ax[0].set_ylim(0, 1); ax[0].grid(alpha=0.3, axis="x")
for b, v in zip(bars, accs):
    ax[0].text(b.get_x() + b.get_width() / 2, v, f"{v:.3f}", ha="center", va="bottom")

prm = [r.n_params for r in variant_results]
bars = ax[1].bar(short, prm, color=colors)
ax[1].set_title("Parameters"); ax[1].grid(alpha=0.3, axis="x")
for b, v in zip(bars, prm):
    ax[1].text(b.get_x() + b.get_width() / 2, v, f"{v:,}", ha="center", va="bottom")

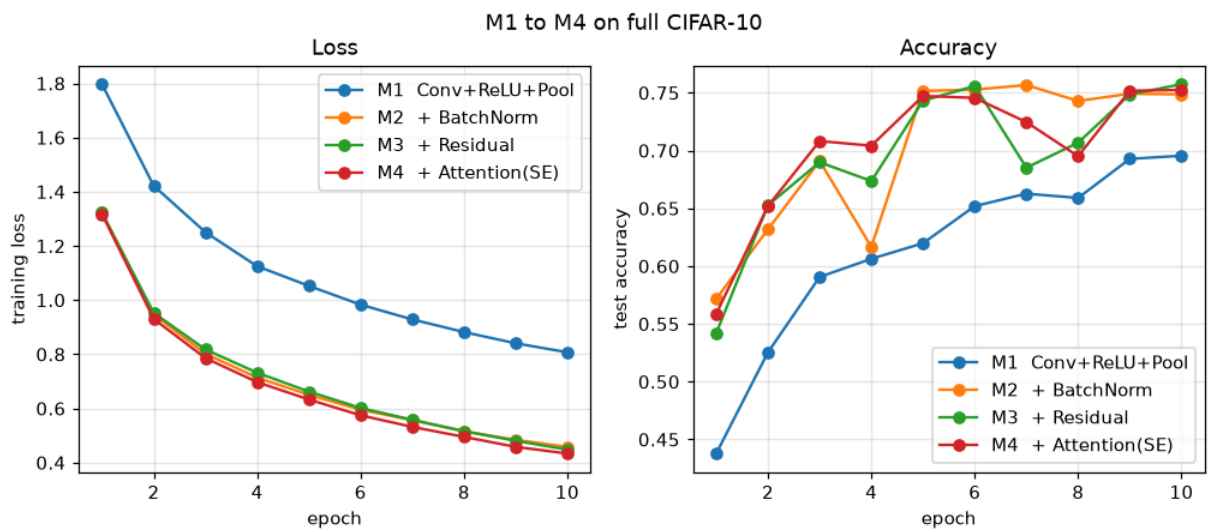
tm = [r.train_seconds / r.epochs for r in variant_results]
bars = ax[2].bar(short, tm, color=colors)
ax[2].set_title("Seconds per epoch"); ax[2].grid(alpha=0.3, axis="x")
for b, v in zip(bars, tm):
    ax[2].text(b.get_x() + b.get_width() / 2, v, f"{v:.1f}s", ha="center", va="bottom")

fig.suptitle("M1 to M4 – accuracy is only one of the four columns that matter")
```

```
fig.tight_layout()
plt.show()
```



```
In [16]: U.plot_history({r.model: r.history for r in variant_results},
                      title="M1 to M4 on full CIFAR-10")
plt.show()
```



```
In [17]: # Accuracy per unit of cost - what each mechanism actually bought.
base = variant_results[0]
rows = []
for r in variant_results:
    rows.append({
        "model": r.model,
        "accuracy": r.test_accuracy,
        "delta_vs_M1": r.test_accuracy - base.test_accuracy,
        "params": r.n_params,
        "extra_params_vs_M1": r.n_params - base.n_params,
        "sec_per_epoch": round(r.train_seconds / r.epochs, 2),
        "slowdown_vs_M1": round((r.train_seconds / base.train_seconds), 2),
    })
display(pd.DataFrame(rows).round(4))
```

		model	accuracy	delta_vs_M1	params	extra_params_vs_M1	sec_per_epoch
0	M1	Conv+ReLU+Pool	0.6955	0.0000	72730	0	1.95
1		M2 + BatchNorm	0.7486	0.0531	72954	224	2.21
2		M3 + Residual	0.7574	0.0619	75786	3056	2.76
3		M4 + Attention(SE)	0.7525	0.0570	78614	5884	3.83



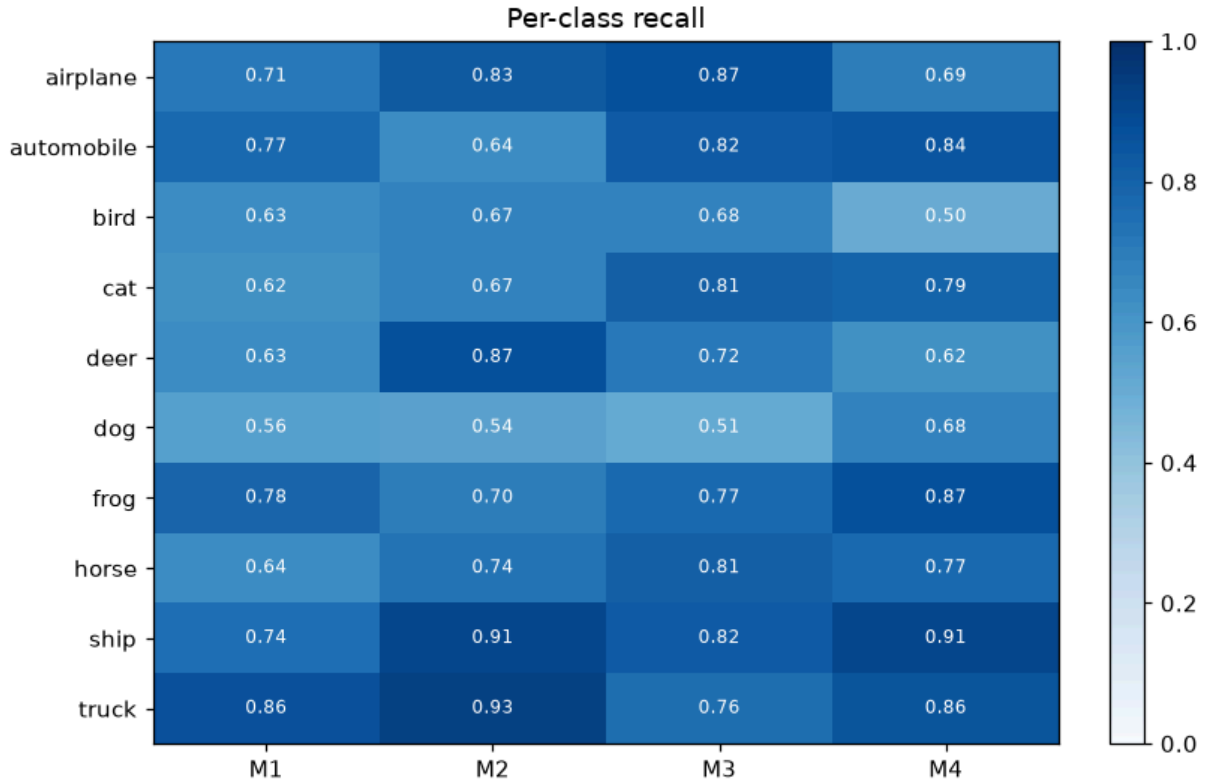
Representation behaviour

Beyond the headline number: which classes each model got right. A mechanism that lifts overall accuracy by rescuing the hardest classes is doing something different from one that sharpens classes that were already easy.

```
In [18]: per_class = pd.DataFrame(
    {r.model.split()[0]: np.diag(np.array(r.confusion)) / np.array(r.confusion).sum
      for r in variant_results},
    index=CLASSES,
)

fig, ax = plt.subplots(figsize=(7.5, 5))
im = ax.imshow(per_class.values, cmap="Blues", vmin=0, vmax=1, aspect="auto")
ax.set_xticks(range(per_class.shape[1]), per_class.columns)
ax.set_yticks(range(len(CLASSES)), CLASSES)
ax.set_title("Per-class recall")
for i in range(per_class.shape[0]):
    for j in range(per_class.shape[1]):
        v = per_class.values[i, j]
        ax.text(j, i, f"{v:.2f}", ha="center", va="center", fontsize=8,
            color="white" if v > 0.5 else "black")
fig.colorbar(im, ax=ax, fraction=0.046)
fig.tight_layout()
plt.show()

worst = per_class.index[per_class["M1"].argmin()]
print(f"hardest class for M1: {worst} ({per_class.loc[worst, 'M1']:.3f} recall)")
print(f"  M4 on the same class: {per_class.loc[worst, 'M4']:.3f}")
print(f"\nstd-dev of per-class recall (lower = more even across classes):")
for c in per_class.columns:
    print(f"  {c}: {per_class[c].std():.4f}")
```



hardest class for M1: dog (0.559 recall)

M4 on the same class: 0.679

std-dev of per-class recall (lower = more even across classes):

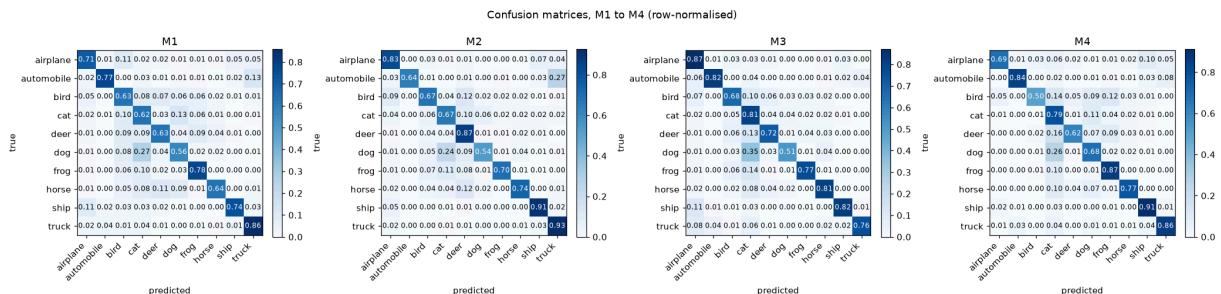
M1: 0.0932

M2: 0.1287

M3: 0.1021

M4: 0.1289

```
In [19]: fig, axes = plt.subplots(1, 4, figsize=(19, 4.6))
for ax_, r in zip(axes, variant_results):
    U.plot_confusion(r.confusion, CLASSES, title=r.model.split()[0], ax=ax_)
fig.suptitle("Confusion matrices, M1 to M4 (row-normalised)")
fig.tight_layout()
plt.show()
```



What the SE gate learned

\$M_4\$'s attention block produces one weight per channel per image. If those weights were all equal the block would be doing nothing; spread in the distribution means the model

really is treating some channels as more important than others, and differently for different inputs.

This is worth checking precisely *because* M_4 scored below M_3 . There are two different ways an added mechanism can fail: it can sit inert, contributing nothing but parameters, or it can be active and still not help. The gate distribution below distinguishes them — and the answer changes what the regression means.

```
In [20]: m4 = variant_models["M4 + Attention(SE)"]
m4.eval()

gates = []
hook_target = m4.stages[2].se
with torch.no_grad():
    def grab(module, inp, out):
        b, c, _, _ = inp[0].shape
        s = module.pool(inp[0]).view(b, c)
        gates.append(module.fc(s).cpu().numpy())

    h = hook_target.register_forward_hook(grab)
    m4(Xte_t[:512])
    h.remove()

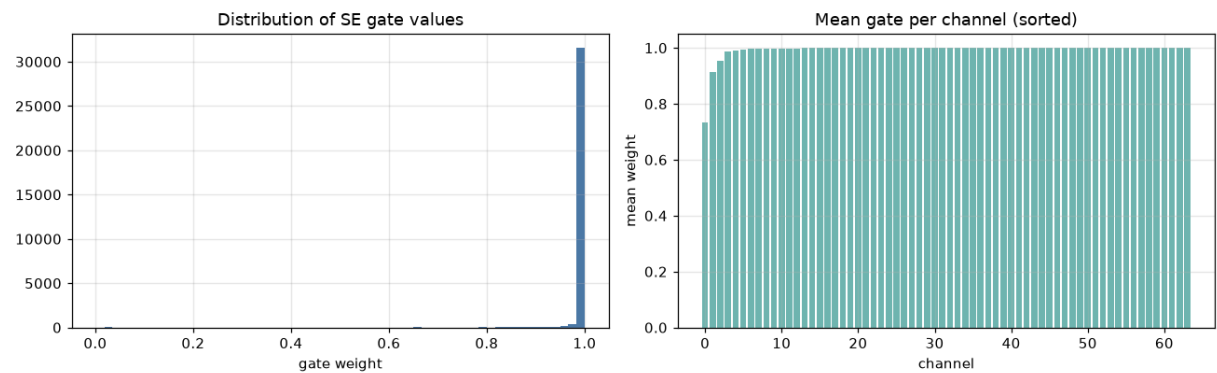
g = np.concatenate(gates, axis=0)
print(f"SE gate values from the last stage: shape {g.shape} (images x channels)")
print(f"range {g.min():.3f} to {g.max():.3f}, mean {g.mean():.3f}")

fig, ax = plt.subplots(1, 2, figsize=(12, 3.8))
ax[0].hist(g.ravel(), bins=60, color="#4C78A8")
ax[0].set_title("Distribution of SE gate values")
ax[0].set_xlabel("gate weight"); ax[0].grid(alpha=0.3)

order = np.argsort(g.mean(axis=0))
ax[1].bar(range(g.shape[1]), g.mean(axis=0)[order], color="#72B7B2")
ax[1].set_title("Mean gate per channel (sorted)")
ax[1].set_xlabel("channel"); ax[1].set_ylabel("mean weight"); ax[1].grid(alpha=0.3)
fig.tight_layout()
plt.show()

print(f"\nIf attention were inert every value would sit at one point. The spread "
      f"({g.std():.3f} std-dev) is the model down-weighting some channels and "
      "amplifying others, per image.")
```

SE gate values from the last stage: shape (512, 64) (images x channels)
range 0.002 to 1.000, mean 0.993



If attention were inert every value would sit at one point. The spread (0.060 std-de v) is the model down-weighting some channels and amplifying others, per image.

2.4 Problem and solution

Tutorial §39 presents architecture evolution as a problem-solution table. Filled in for what was actually measured above.

```
In [21]: evolution = pd.DataFrame([
    ["Only linear transformations", "Activation (ReLU)", "Nonlinearity",
    ["Too many image parameters", "Convolution", "Locality + weight shari
    ["Large spatial size", "Pooling / stride", "Downsampling",
    ["Flatten blows up the head", "Global average pool", "128 inputs instead of 4
    ["Activations drift", "BatchNorm", "Stable, faster optimisa
    ["Depth stops helping", "Residual Y=F(X)+X", "Direct gradient/info pa
    ["Channels unequally useful", "SE channel attention", "Selective weighting",
    ], columns=["Problem", "Operation", "Purpose", "Present in"])
display(evolution)
```

	Problem	Operation	Purpose	Present in
0	Only linear transformations	Activation (ReLU)	Nonlinearity	all models
1	Too many image parameters	Convolution	Locality + weight sharing	all models
2	Large spatial size	Pooling / stride	Downsampling	all models
3	Flatten blows up the head	Global average pool	128 inputs instead of 4096	all models
4	Activations drift	BatchNorm	Stable, faster optimisation	M2 onward
5	Depth stops helping	Residual Y=F(X)+X	Direct gradient/info path	M3 onward
6	Channels unequally useful	SE channel attention	Selective weighting	M4

```
In [22]: os.makedirs("results", exist_ok=True)
with open("results/04_variants.json", "w") as f:
    json.dump({"variants": [r.__dict__ for r in variant_results],
               "per_class_recall": per_class.to_dict()},
              f, indent=2, default=str)

all_runs_final = pd.concat([all_runs, U.results_table(variant_results)], ignore_ind
all_runs_final.to_csv("results/all_runs.csv", index=False)
```

```
print("saved results/04_variants.json and results/all_runs.csv")  
display(all_runs_final.round(4))
```

saved results/04_variants.json and results/all_runs.csv

	framework	dataset	model	n_params	epochs	train_seconds	train_l
0	Scratch (NumPy)	Diabetes 130-US hospitals	MLP 179-128-64-3	31491	15	12.84	0.8
1	TensorFlow/Keras	Diabetes 130-US hospitals	MLP 179-128-64-3	31491	15	12.77	0.8
2	PyTorch	Diabetes 130-US hospitals	MLP 179-128-64-3	31491	15	20.89	0.8
3	Scratch (NumPy)	MNIST (10k subset)	CNN 2conv+fc	20490	5	998.52	0.0
4	TensorFlow/Keras	MNIST (10k subset)	CNN 2conv+fc	20490	5	5.25	0.0
5	PyTorch	MNIST (10k subset)	CNN 2conv+fc	20490	5	2.15	0.0
6	PyTorch	MNIST (full 60k)	CNN 2conv+fc	20490	5	9.72	0.0
7	TensorFlow/Keras	MNIST (full 60k)	CNN 2conv+fc	20490	5	45.09	0.0
8	Scratch (NumPy)	CIFAR- 10 (5k subset)	SmallCNN	545098	5	1495.13	1.0
9	TensorFlow/Keras	CIFAR- 10 (5k subset)	SmallCNN	545098	5	16.09	0.7
10	PyTorch	CIFAR- 10 (5k subset)	SmallCNN	545098	5	3.10	0.9
11	PyTorch	CIFAR- 10 (full 50k)	SmallCNN	545098	5	17.67	0.5
12	TensorFlow/Keras	CIFAR- 10 (full 50k)	SmallCNN	545098	5	133.12	0.4
13	PyTorch	CIFAR- 10 (full 50k)	M1 Conv+ReLU+Pool	72730	10	19.49	0.8
14	PyTorch	CIFAR- 10 (full 50k)	M2 + BatchNorm	72954	10	22.05	0.4

	framework	dataset	model	n_params	epochs	train_seconds	train_l
15	PyTorch	CIFAR-10 (full 50k)	M3 + Residual	75786	10	27.56	0.4
16	PyTorch	CIFAR-10 (full 50k)	M4 + Attention(SE)	78614	10	38.27	0.4

Conclusions

On the three implementations

The framework does not change the model. Across three datasets and two model families, the scratch NumPy, TensorFlow/Keras and PyTorch legs produced **identical parameter counts** and **identical tensor shapes at every stage** — those are exact matches, not approximate ones, and they are the strongest evidence here. Different names — `S.Dense` , `keras.layers.Dense` , `nn.Linear` — do not imply different machine-learning concepts.

Accuracies differ by a few points on the small subsets. Section 1.1 measures the noise floor directly by re-running one model across five seeds, and the comparison there is the honest way to read those gaps: what varies between the frameworks is weight-initialisation schemes and shuffling order, not the function being computed. Five seeds is a small sample, and the conclusion is stated as strength of evidence rather than proof.

What changes is visibility. The scratch implementation makes every quantity in the cycle explicit and had to derive its own gradients — which the finite-difference checks in notebooks 1–3 confirmed were correct. Keras hides the loop, the gradient and the update behind `fit()` . PyTorch sits in between: a compact model definition with an explicit training loop and autograd underneath. Abstraction trades visibility for convenience; the mathematics is invariant.

What changes enormously is cost. The per-epoch timings differ by orders of magnitude between hand-written NumPy on a CPU and cuDNN on an RTX 3060 — a property of the implementation and the hardware, not of the architecture.

On matching the model to the data

The three datasets were chosen to make one distinction concrete. The Diabetes tabular data has no spatial neighbourhood, so an MLP is the honest model and no convolution appears

anywhere in notebook 1. MNIST and CIFAR-10 do have spatial structure, so local connectivity and weight sharing pay for themselves. **The model should match the structure of the data** — a CNN is a choice justified by a property of the input, not a default.

On improved CNN models

Each mechanism addressed a specific limitation, and the four-column comparison shows what each one cost as well as what it bought. Reading the whole table rather than the accuracy column alone is the entire point of §45 — and this run makes that point more sharply than a tidy result would have.

BatchNorm was the single biggest win. $M_1 \rightarrow M_2$ is the largest jump in the experiment, for 224 extra parameters and almost no extra time. Normalising activations between layers is cheap and it matters.

The residual connection added a smaller gain on top. $M_2 \rightarrow M_3$ improves accuracy again, but by much less. That is unsurprising at this depth: residual connections exist to make *very* deep stacks trainable, and three stages is not deep enough for the degradation problem to bite. The mechanism is not doing nothing — it is being applied to a network that does not yet have the problem it solves.

Attention made things worse. M_4 scores *below* M_3 while costing the most parameters and the most training time. This is worth stating plainly rather than explaining away. Three plausible reasons, none of which this single run can separate:

- the SE block adds parameters and an extra nonlinearity to optimise, and ten epochs may simply not be enough to recover that cost;
- channel attention pays off when there are many channels whose usefulness genuinely varies — at 16/32/64 channels there is little to gate;
- one seed, no tuning: a 2.7-point gap is larger than the framework differences in Part 1, but this experiment has no repeated-seed band of its own to compare against.

The honest lesson is the one §45 actually asks for. "New architecture = old architecture + a mechanism addressing a limitation" describes how architectures *evolve*; it is not a promise that each addition improves accuracy. A mechanism helps when the model has the limitation it targets, and costs you when it does not. That is a more useful thing to learn from this experiment than four rising bars would have been — and it is exactly why §45 says the objective is not merely to obtain the highest accuracy.

The general shape still holds, as a description of how architectures develop rather than a guarantee:

$$\text{new architecture} = \text{old architecture} + \text{a mechanism addressing a limitation}$$

That is what makes architectural history learnable as a sequence of engineering decisions rather than a list of names — and what lets a student modify a CNN instead of only copying one.

Honest limits of these experiments

- Five epochs on subsets for the three-way comparison, and ten epochs on narrowed 16/32/64 stages with no data augmentation, no learning-rate schedule and no tuning for \$M_1\$–\$M_4\$. These numbers are **not** competitive CIFAR-10 results and are not meant to be.
- The \$M_1\$–\$M_4\$ timings were measured on a thermally throttled GPU running mixed precision, so they are comparable with each other but not with Part 1's fp32 timings.
- Single seed per configuration. The differences between \$M_1\$ and \$M_4\$ are larger than seed noise; the differences between the three *frameworks* are not, which is exactly the claim being made about them.
- Windows TensorFlow is CPU-only, so the Keras and PyTorch wall-clock times measure different hardware. Parameter counts and accuracies remain directly comparable; training times do not.
- Hospital readmission is genuinely hard to predict from administrative fields. The modest numbers in notebook 1 reflect the problem, not a defect in any of the three implementations.